

CLASS 10 · ICSE · SELINA CONCISE MATHEMATICS · PYQ SOLUTIONS

Chapter 22: Heights and Distances

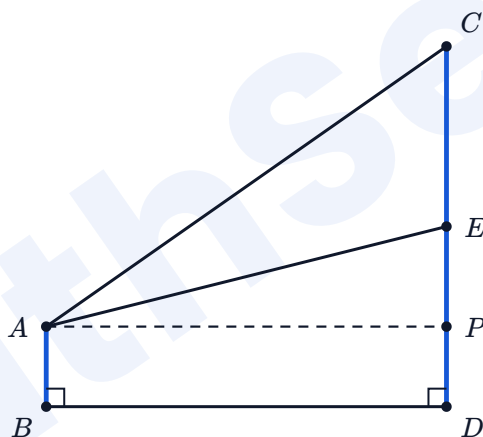
Step-by-step solutions to the previous year questions

The question numbers match the PYQ question sheet.

1 Problems with two triangles

1. A vertical tower AB stands 100 m from the foot D of a building CD . E is the mid-point of CD . The angle of depression of the top A of the tower is 35° from C and 14° from E . Using tangent values rounded to two decimal places, find the height of (a) the tower (b) the building.

ICSE 2025, 5 marks



SOLUTION

Draw AP horizontal, meeting CD at P . Then $AP = BD = 100$ m and $PD = AB$. The angles of depression equal $\angle CAP = 35^\circ$ and $\angle EAP = 14^\circ$ (alternate angles).

From the tables, $\tan 35^\circ = 0.70$ and $\tan 14^\circ = 0.25$.

$CP = 100 \tan 35^\circ = 70$ m and $EP = 100 \tan 14^\circ = 25$ m, so $CE = 70 - 25 = 45$ m.

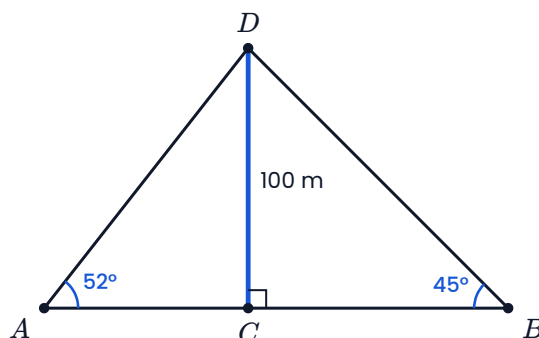
E is the mid-point of CD , so $CD = 2 \times 45 = 90$ m and $ED = 45$ m.

$AB = PD = ED - EP = 45 - 25 = 20$ m

Answer: (a) tower 20 m (b) building 90 m

2. A tree CD is 100 m high. From two points A and B on the ground on opposite sides of the tree, its top D is seen at angles of elevation of 52° and 45° . Find the distance AB to the nearest metre.

ICSE 2024, 5 marks



SOLUTION

In $\triangle ACD$: $\tan 52^\circ = \frac{100}{AC}$. From the tables $\tan 52^\circ = 1.2799$, so $AC = \frac{100}{1.2799} \approx 78.13$ m.

In $\triangle BCD$: $\tan 45^\circ = \frac{100}{CB}$, so $CB = 100$ m.

$$AB = AC + CB = 78.13 + 100 = 178.13$$

Answer: $AB \approx 178$ m

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